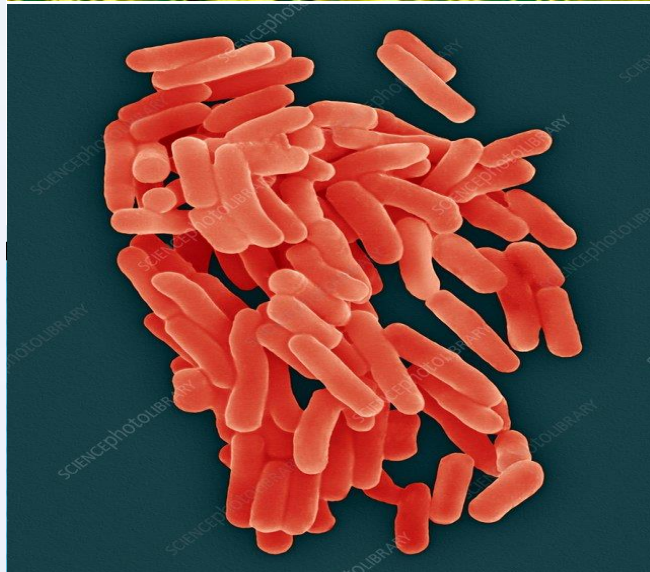
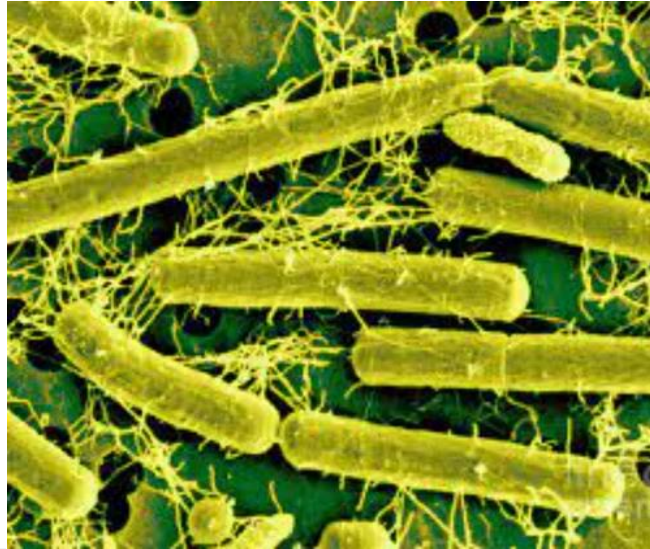


BIOMINING



Mining

- Mining is the extraction of valuable geological materials and minerals from the surface of the Earth.
- **Types of mining:**
 1. **Underground mines:** when ore materials lies a considerable distance below the surface, underground techniques considered for its extraction.
 2. **Opencast mines:** Open-pit mining, also known as opencast mining, is a surface mining technique that extracts minerals from an open pit in the ground.
- Open-pit mining is the most common method used throughout the world for mineral mining and does not require extractive methods or tunnels

Types of mining

Underground mines

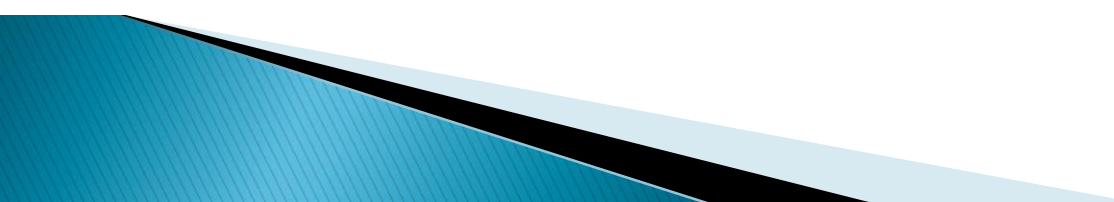


Opencast mines



Impact of mining on the Environment

- Due to mining, environment is deteriorating day by day as well as the indigenous people are losing their original habitat, source of food, drinking water and shelter.
 - Ground water and surface water sources near the mining area is contaminated through leaching and over flow of overburden dumps especially during rainy season.
 - Direct discharge of untreated or improperly treated mining effluents (contain various poisonous elements including heavy metals), into the surface water can be harmful for the aquatic animal like fish.
 - It causes serious health hazards to human being who consume this contaminated water.
- 

- Conventional mining technologies use physical-chemical processes which need large amounts of energy for roasting/smelting and it also produce harmful gases such as sulfur dioxide.
 - Removal of large areas of topsoil can destroy habitats thus disturbing ecosystem.
 - These process also affect the atmosphere through carbon emission which contributes to climate change.
 - Coal dust, or other hazardous particles produced during mining process, inhaled by workers causes various health effect like respiratory problem.
 - Ecofriendly process called biomining is used for extraction of metals from low grade ore.
- 

Why Bio mining?

- High-grade ore contains a large concentration of the desired mineral. It has less impurities.
- Low-grade ore has a smaller concentration of valuable minerals or metals. It has more impurities.
- High grade ores is extracted by using conventional mining process including, smelting, roasting etc.
- Low grade ores are not suitable for direct smelting. So metal can be extracted from low grade ore using bio mining or using the activity of microorganisms.

Why Biomining?

Pollution

Dust
Gases
Toxic compounds

1. Cheap
2. Reliable
3. Efficient
4. Safe
5. Environmentally friendly

Biomining or mineral leaching

- It is the extraction of metals from their ores by using micro-organisms such as Bacteria/ fungi.
- **It is an effective and green technology to mine metals.**
- This techniques may also be used to clean up sites that have been polluted with metals.
- Bio-mining is clearing landfills by converting waste into compost, methane gas, bio-diesel and power.
- Bio-mining of the garbage mounds, **make manure from the solid waste.**
- Most current bio-mining operations target **valuable metals like copper, uranium, nickel, and gold** that are **commonly found in sulfidic (sulfur-bearing) minerals.**



Figure: 27 Garbage bio-mining

Introduction

- Biomining is the broad term that describes the extraction of specific metals from their ores through biological means, usually microorganisms.
- It is an alternative to more traditional physical-chemical methods of mineral processing.
- The application of biomining processes predates by centuries the understanding of the role of microorganisms in metal extraction.
- However, the modern era of biomining began with the discovery of the bacterium *Thiobacillus ferrooxidans*.

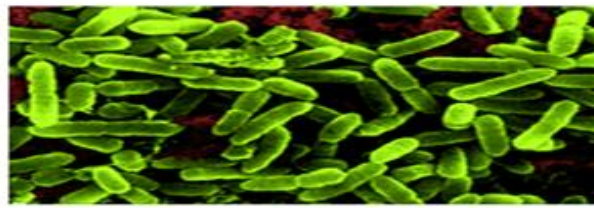


Fig.: Microscopic image of *Thiobacillus ferrooxidans*

- **Bio-mining encompasses two processes: Bioleaching and Bio oxidation**

1. Bioleaching or Bio mining

- The ability of microorganism to solubilized metals from ore is called **Bioleaching**.
- Bioleaching is the process by which metals are dissolved from ore bearing rocks using microorganism.
- Bioleaching is the conversion of an insoluble metal (CuS, NiS, and ZnS) into a soluble form (CuSO₄, NiSO₄ and ZnSO₄).
- Bioleaching is also called microbial leaching.

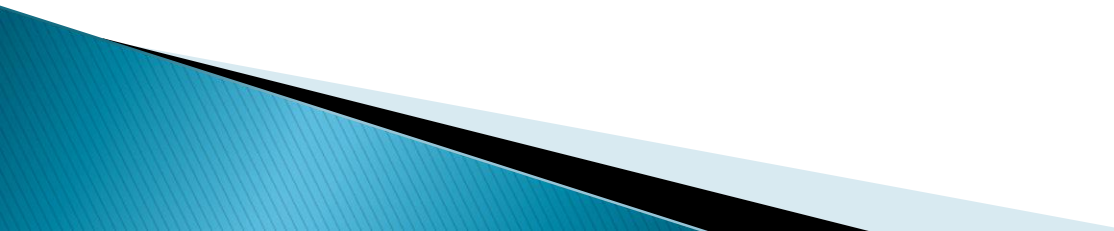


Ore

Microorganism

Pure metal

2.Bio oxidation

- Bio oxidation is a process by which the recovery of metal is enhanced by microbial decomposition of the mineral, but the target material is not necessarily solubilized.
 - In Bio oxidation undesired metal along with impurities or other compounds are leached away from the ore.
 - In bioleaching the desired metal is leached from the ore.
- 

Rio Tinto mining

- Microbes played a important role in the extraction of Cu from Rio Tinto mine, Spain.
- The Rio Tinto (Red River) river is **appeared red** because of the **presence of high concentration of ferric iron** due to natural microbial activity.



How Bio mining works?

The Rio Tinto acidic river, Spain

Chalcopyrite
(CuFeS_2)



Microbes

Sulfide-consuming bacteria
separate Cu and Fe from
ore (CuFeS_2)

Iron and S is
consumed by
microbes



The Rio Tinto acidic river ($\pm \text{pH } 2.3$)

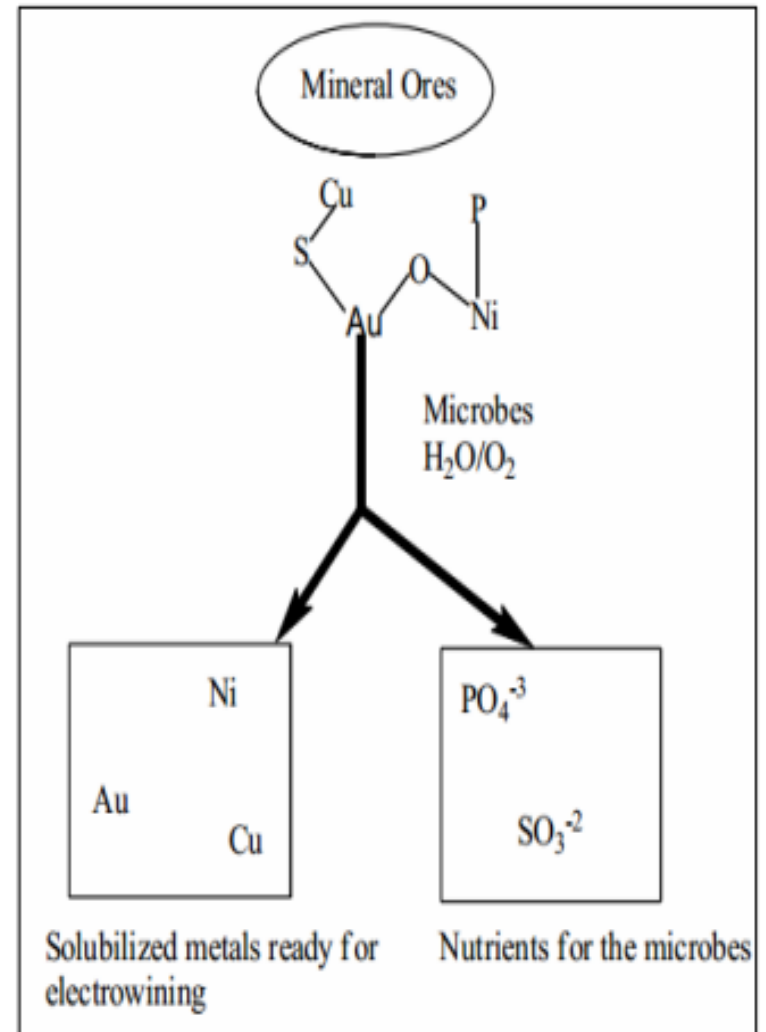


Figure: 2 A Schematic of Biomining

Microorganism used in Bioleaching

The most commonly used microorganism in bioleaching are:

- Thiobacillus thiooxidans
- Thiobacillus ferrooxidans

Other microorganism:

- Sulfolobus-Cu, As
- Saccharomyces cerevisiae-Cu, Pb, Sn
- Penicillium simplicissimum-Cr
- Certain fungi Aspergillus niger used to extract Cu and Ni.
- Aspergillus oryzae used for Au.

Chemistry of Bioleaching

- *Thiobacillus thiooxidans* and *T. ferrooxidans* have always been found to be present on the leaching dump.
- The species of *thiobacillus* is most extensively studied gram –ve bacteria which derives energy from oxidation of Fe^{2+} .
- The reaction mechanisms are of two types, i.e.,
 - Direct bacterial leaching (Contact leaching)
 - Indirect bacterial leaching

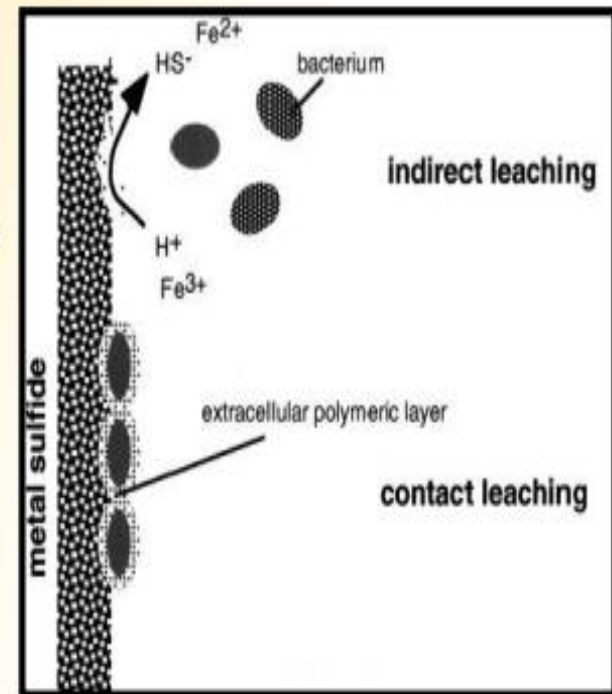
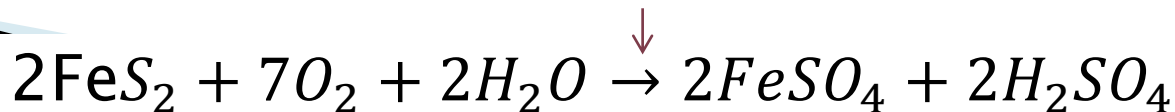


Fig.: Direct and indirect bioleaching

Direct Bacterial Leaching

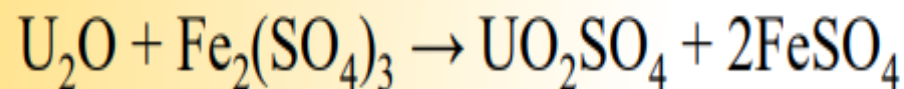
- Enzymatic attack.
- In this process, a physical contact exist between bacteria and ores and oxidation of mineral takes place through enzymatically catalyzed steps.
- Microorganism can oxidize metal sulfides by a direct mechanism obtaining electrons directly from the reduced minerals.
- And convert Fe^{+2} to Fe^{+3} .
- Example of pyrite is oxidized to ferric sulphate.

Thiobacillus



Indirect bacterial leaching

- In this process the microbes are not in direct contact with minerals, but bacteria produces strong oxidising agents such as ferric ion and sulfuric acid on oxidation of soluble iron or soluble sulfur respectively.
- For indirect bioleaching, acidic environment is absolutely essential in order to keep ferric iron and other metals in solution.
- Acidic environment maintained by oxidation of iron, sulfur, metal sulphides or by dissolution of carbonate ions.
- Example: Bioleaching of uranium



Direct bioleaching:

Direct enzymatic attack on the minerals by microorganisms. Certain bacteria (e.g., *T. ferrooxidans*) can transfer electrons (coupled with ATP) from iron or sulfur to oxygen.

These organisms can obtain energy from the oxidation of Fe^{2+} to Fe^{3+} or from the oxidation of sulfur to sulfate.

Indirect bioleaching:

Bacteria produce strong oxidizing agents (ferric iron and sulfuric acid) helps in oxidation of soluble iron or soluble sulfur respectively.

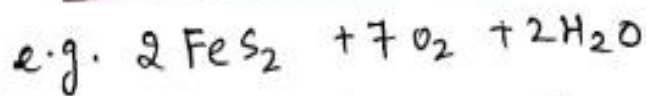
Ferric iron or sulfuric acid, being powerful oxidizing agents react with metals and extract them.

Mechanism of Bio mining

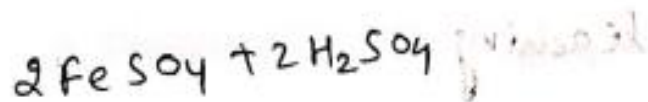
Direct



- In this process, there is a direct enzymatic attack on the minerals by M.O.
- M.O. can oxidize Metal sulfides by a direct Mechanism obtaining e^- directly from reduced minerals.



↓ Thiobacillus



Indirect Method.

- No need of direct physical contact b/w M.O. & Mineral ore.
- Iron act as a e^- carrier. It obtain e^- from Mineral & transfer to M.O.
- Bacteria produce strong oxidizing agents such as ferric iron, sulfuric acid on oxidation of soluble Fe/S.
- Ferric Iron / sulfuric acid being powerful oxidizing agents react with mineral metals & extract them.

Bioleaching



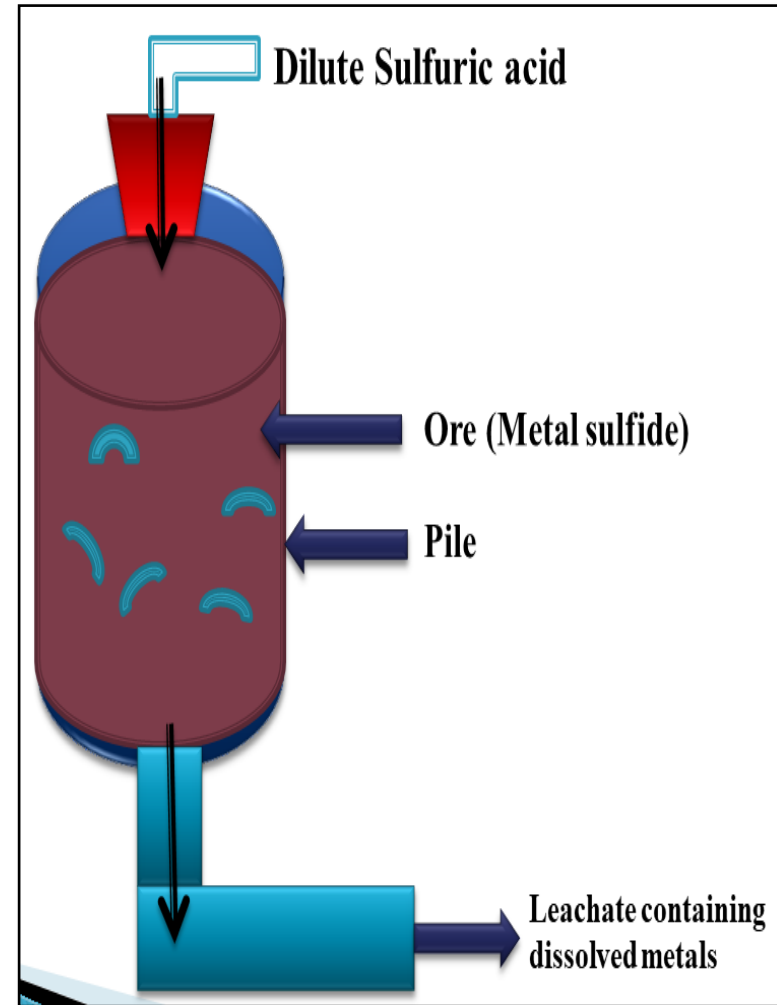
- ▶ Extraction of metals and ores using principle components of water, air and microorganisms which are readily found within the environment.
 - ▶ It directly dissolves metal of interest.
 - ▶ It involves the following processes
- 

Commercial process of Bioleaching

- The naturally occurring mineral leaching is very slow.
- Microbial bioleaching process can be optimized by creating ideal conditions such as pH, temperature, nutrients, CO₂ and O₂ concentration.
- Low grade ore is dumped in a large pile (the leach dump) and the desired microorganism with nutrient, sulfuric acid solution (pH 2) etc. is percolated down through the pile.
- The microorganism grow and produce more acid.
- It dissolved more metals (dissolved solid metal attached to ore into liquid metal).

Commercial Process of Bioleaching

- The liquid coming out at the bottom of the pile, rich in the mineral is collected and transported to a precipitation plant where the metal is reprecipitated and purified.
- The liquid (leach liquor) is then pumped back to the top of the pile and the cycle is repeated for further metal extraction.
- The most important microbes that are used include iron and sulphur oxidizing chemolithotrophs.



Types of Bioleaching

- There are three commercial processes used in bioleaching:

1. Slope leaching
2. Heap leaching
3. In- situ leaching

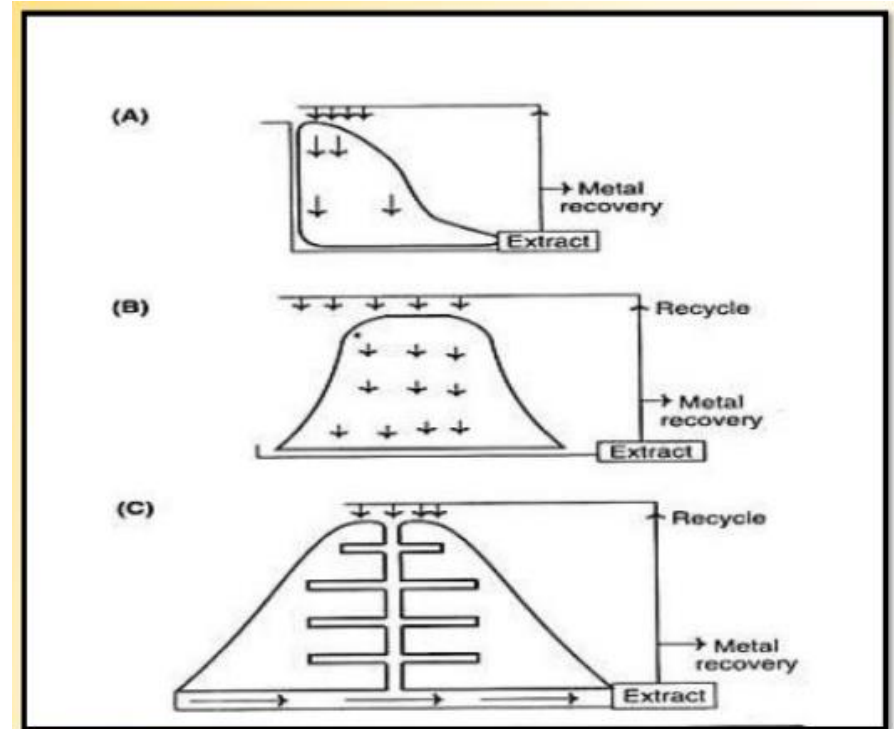
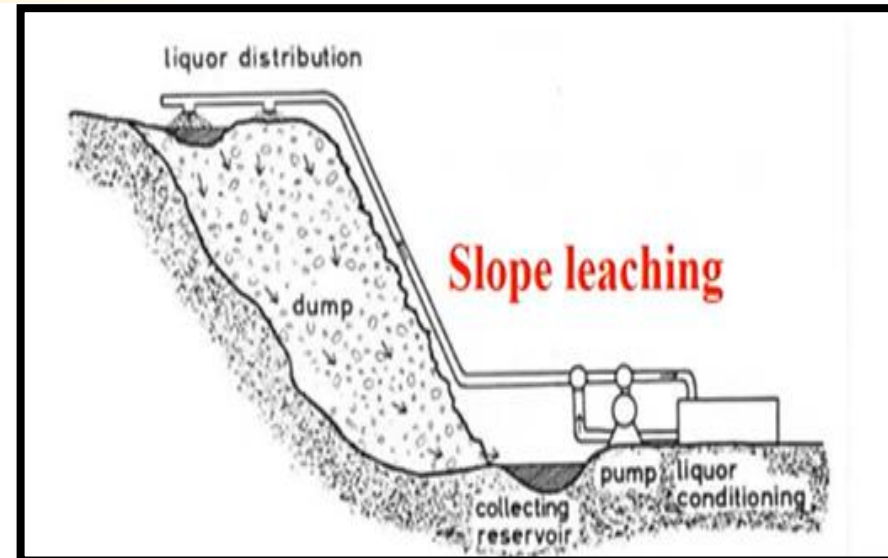


Fig.: Commercial bioleaching processes- (A) Slope leaching, (B) Heap leaching and (C) In-situ leaching

Slope leaching

- Here the ores are first ground to get fine pieces and then dumped into large leaching dump.
- Water containing inoculum of *thiobacillus* is continuously sprinkled over the ore.
- Water is collected from the bottom and used to extract metals and generate bacteria in an oxidation pond.



Heap Leaching or Bioheaps

- The ore is arranged in the form of large Heaps.
- Water containing thiobacillus is continuously sprinkled over the ore.
- Acidic leaching solution is percolated through the crushed ore, and microbes growing on the surface of the mineral in the heap produce the ferric iron and acid that result in mineral dissolution and metal solubilization (bacterial oxidation of ferrous iron and sulfide).
- Metal-containing water is collected from the bottom and used to extract metal.

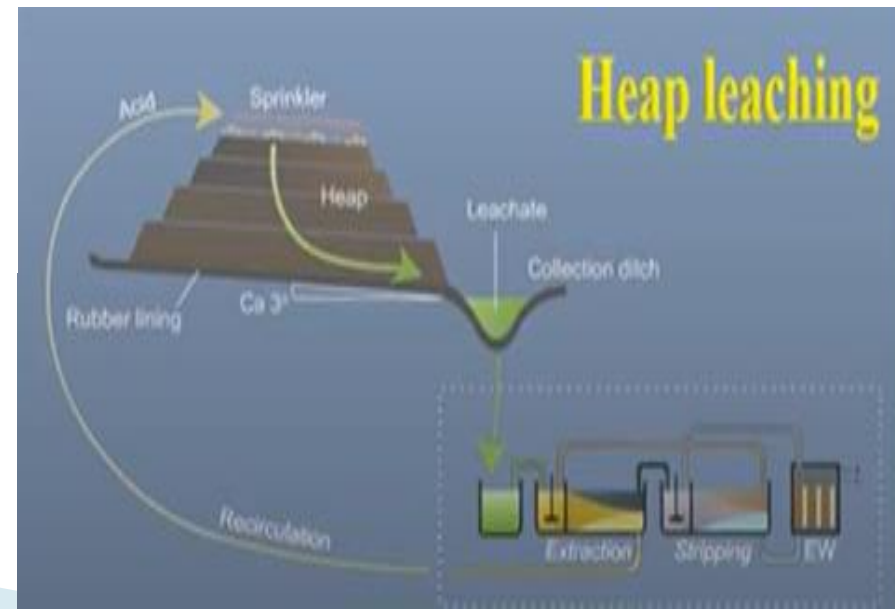
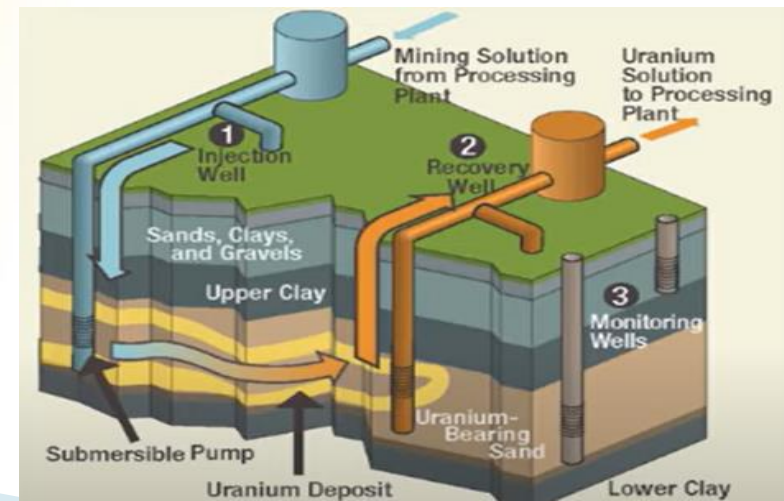




Fig. 3 Photograph of the copper heap-leaching operation at Cerro Colorado, Chile. Heaps of copper-containing mineral have been stacked, an irrigation system has been placed on top of the heap, and air is blown into the heap from below. The heap has been covered with an insulation thermofilm to reduce heat loss. (Photo courtesy of Rodolfo Reyes via Chris du Plessis.)

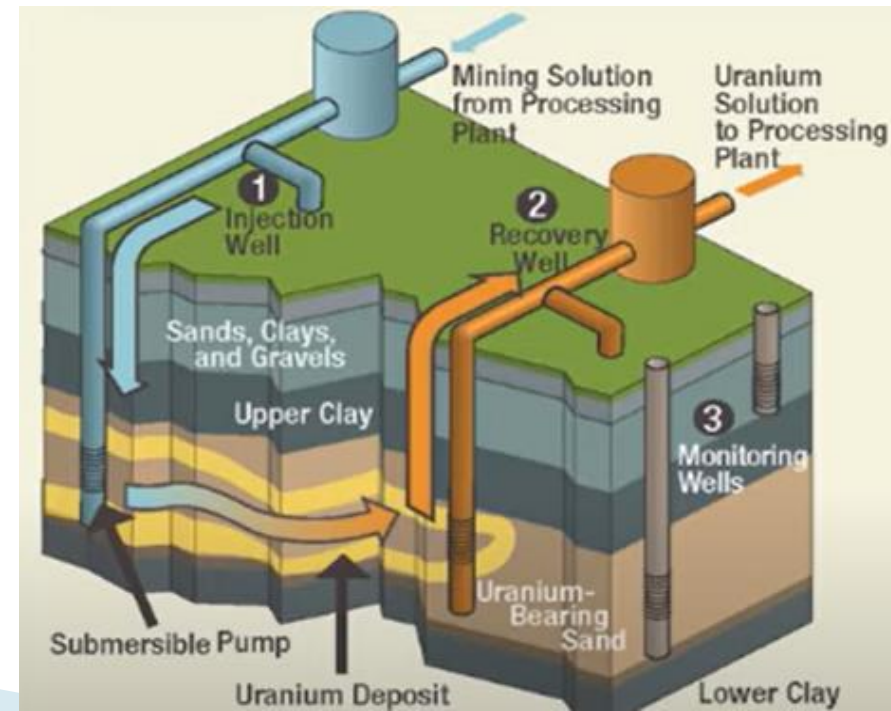
In-situ Bioleaching

- In this process the ore remains in its original position in the earth.
- Surface blasting of earth is done to increase the permeability of water.
- Water containing *thiobacillus* is pumped through drilled passage to the ores.
- Acidic water seeps through the rock and collects at the bottom.
- Again, water is pumped from bottom.
- Mineral is extracted and water is reused after generation of bacteria.



In-Situ Bioleaching

- In this method the mineral is extracted directly from the mine instead of collecting the ore and transferring to an extracting facility away from the site of the mine.
- **The mine is blasted to reduce the ore size and to increase the permeability of water and is then treated with water and acid solution.**
- **Water containing thiobacillus is pumped through drilled passage to the ores.**
- Acidic water seeps through the rock and collect at the bottom.
- Again water is pumped from bottom.
- Mineral is extracted and water is reused after generation of bacteria.



Factors Affecting Bioleaching

Factors

Effects

1	Choice of bacteria	Suitable bacteria that can survive at high temperatures, acid concentrations, high concentrations of heavy metals, remaining active under such circumstances.
2	pH and temperature	Affects leaching rate, microbial growth
3	Population density	High population density tends to increase the leaching rate
4	Metal tolerance	High metal concentration may be toxic to microbes
5	Oxidation reactions and compositions activity	Needs to be low to obtain the fastest leaching rates and to keep ferric ions and metals in solution
6	Surface area	Rate of oxidation by the bacteria increases with reduction in size of the ore and vice-versa.

FACTORS AFFECTING BIOLEACHING

Factors affecting *Thiobacillus ferrooxidans*:

- i. **Temperature** (optimum: 30 – 50°C).
- ii. **pH** (optimum: 2.3 – 2.5).
- iii. **Iron supply** (2-4 g/L of leach liquor oxygen)
- iv. **Ammonia, Nitrogen, Phosphorus and Mg**
- v. **Water**



Bioleaching of Some Metals

i. Copper bioleaching:

Chalcolite - Cu_2S

Covellite - CuS

Chalcopyrite - CuFeS_2

- Chalcopyrite is oxidized as follows:



- Covellite is oxidized to copper sulfate:

▪ Steps of copper bioleaching:

Cu containing ores are crushed & acidified with sulphuric acid



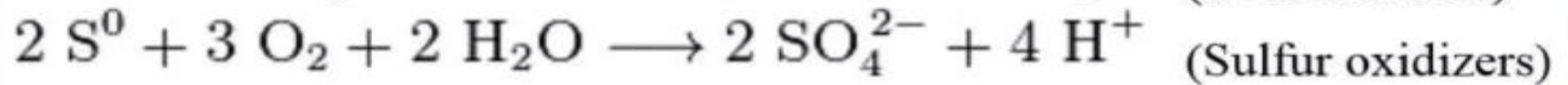
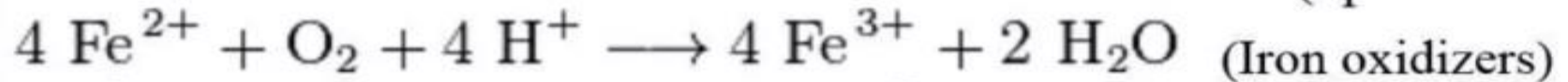
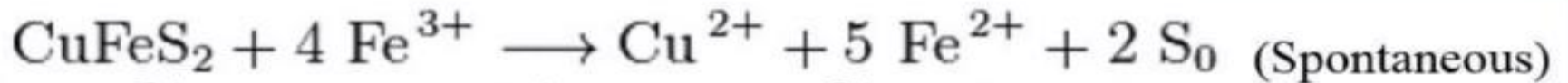
Stacked heaps are irrigated with an iron containing leach liquor
(this leach liquor contains *Thiobacillus ferrooxidans*)



The copper containing solution is removed, the copper is precipitated

Bioleaching of Copper

- Copper Ores: Calcopryrite ($CuFeS_2$), Chalcocite Cu_2S , Covellite CuS
- Copper ore is a low grade ore.
- In bioleaching of Copper, the action of *Acidithiobacillus* involves the oxidation of $CuFeS_2$ via generation of ferric ions.



Net Reaction:



Copper Bioleaching

Bacteria are grown on low grade ores



Biological & chemical processes of Bacteria form a solution rich in Cu ions (leachate)

Cu²⁺ leachate

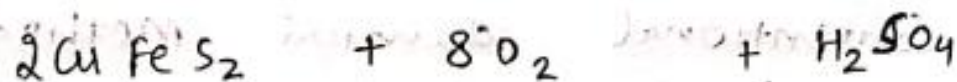


Copper containing solution is removed, the Cu is precipitated.

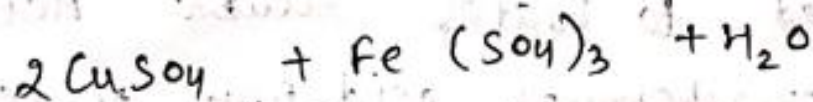
e.g. **chalcopyrite** (contains 26% Cu, 26% Fe, 33% S, 2.5% Zn)

- This is carried out by M.O. *Thiobacillus ferrooxidans* which oxidises insoluble chalcopyrite (CuFeS_2) and converts it into soluble copper sulfate (CuSO_4)

- This is carried out by M.O. *Thiobacillus ferrooxidans* which oxidises insoluble chalcopyrite (CuFeS_2) and converts it into soluble copper sulfate (CuSO_4).
 - sulphuric acid, a byproduct formed in this reaction, maintains acidic env. (low pH) required for growth of M.O.
 - Copper leaching is usually carried out by heap and in situ process.
- ⇒ As the copper containing solution (i.e. copper in dissolved state) comes out, copper can be precipitated and the water is recycled, after adjusting the pH to around 2.



↓ *Thiobacillus*



Benefits of Bioleaching

- Simple process.
- Inexpensive technique.
- No poisonous sulfur dioxide emission as in smelter.
- No need of high pressure and temperature.
- Ideal for low grade sulphide ores.
- Environment friendly process.

Disadvantages of Bioleaching

- Time consuming (takes 6-24 months or longer).
- Have a very low yield of minerals.
- Requires a large open area for treatment
- May have no process control.
- High risk of contamination.
- Inconsistent yield because bacteria can not grow uniformly.

THANK YOU